

Research Assignment 1

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1 Research Project Topic Heading

4. *MW/M31 Dark Matter Halo Major Merger Remnant*

2 Research Question

**Kinematic Properties of the Merged Dark Matter Halos: is there any rotation/what is the evolution of the angular momentum? What is the velocity dispersion profile of the remnant and how does this compare to the original profile of each galaxy?*

OR

Density Profile of the Merged Dark Matter Remnant. What is the density profile of the combined MW + M31 halo? How does this compare to the original profile of each galaxy halo? How does this compare to a Hernquist profile or other dark matter profile (e.g. NFW, Isothermal Sphere)?